Poisson GLM

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1 1. Introduction

This section analyses count data using a Poisson Generalised Linear Model (GLM).
The outcome variable is the number of high-consumption days per household-month (`n_high_days`). As this represents non-negative integer counts over a fixed time period, a Poisson model is an appropriate starting point.

The objective is both explanatory and predictive: to identify which household and weather characteristics are associated with higher rates of high-consumption days, and to quantify these effects in a consistent modelling framework.

2 2. Data & Variables

The analysis is conducted at the household-month level, where daily observations are aggregated to construct the model variables.

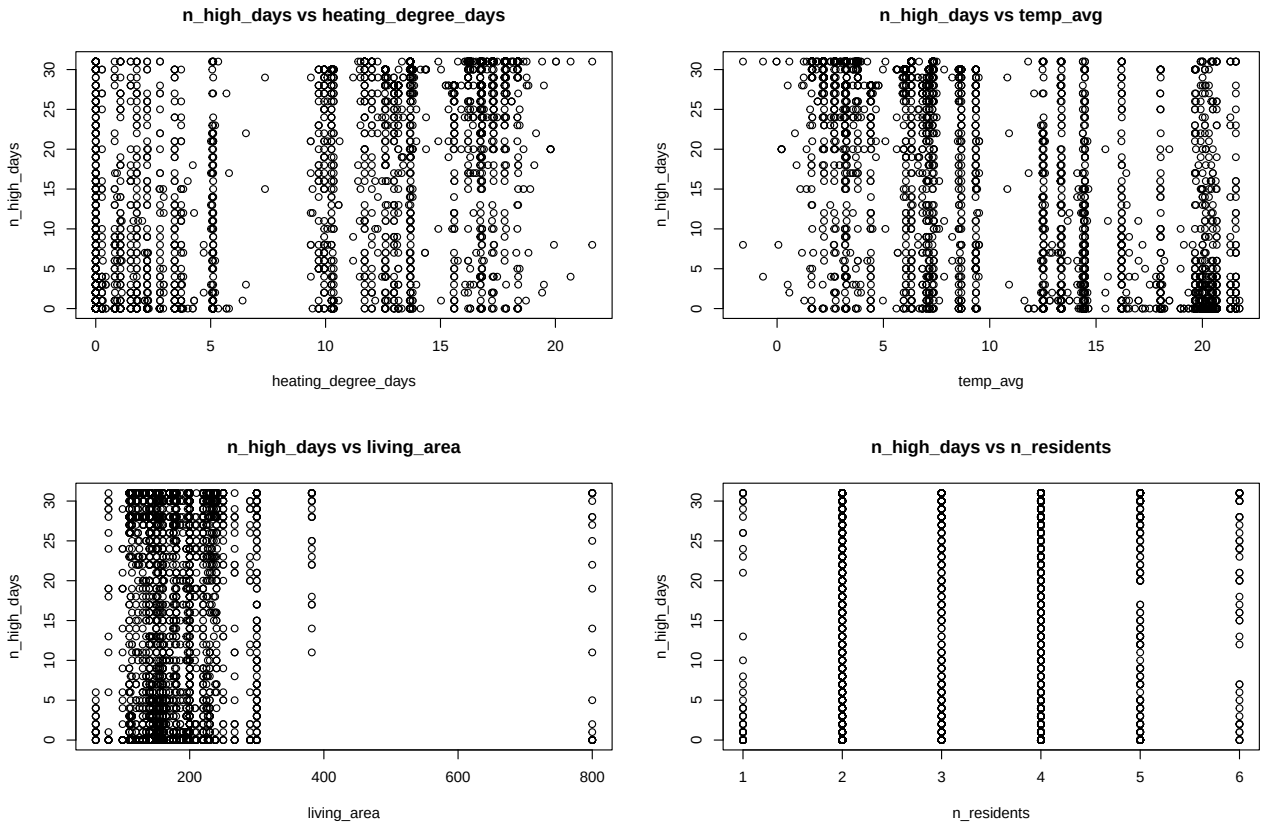
The outcome variable, `n_high_days`, measures the number of days within a household-month with high electricity consumption. To account for differences in observation length, the number of observed days (`n_days`) is included as an offset in the model, allowing the estimation of rates rather than raw counts.

Predictors include weather variables (e.g., temperature and heating degree days), household characteristics (e.g., living area, number of residents), and building-related features (e.g., building type, heat pump type, floor heating). These variables capture key factors expected to influence consumption behaviour.

3 3. Graphical Analysis

A graphical analysis is conducted to examine the distribution of the count outcome and its relationship with the model predictors. The variable `n_high_days` shows typical count-data characteristics, including discreteness, right-skewness, and a substantial number of low-count observations.

3.0.1 Continuous and count predictors



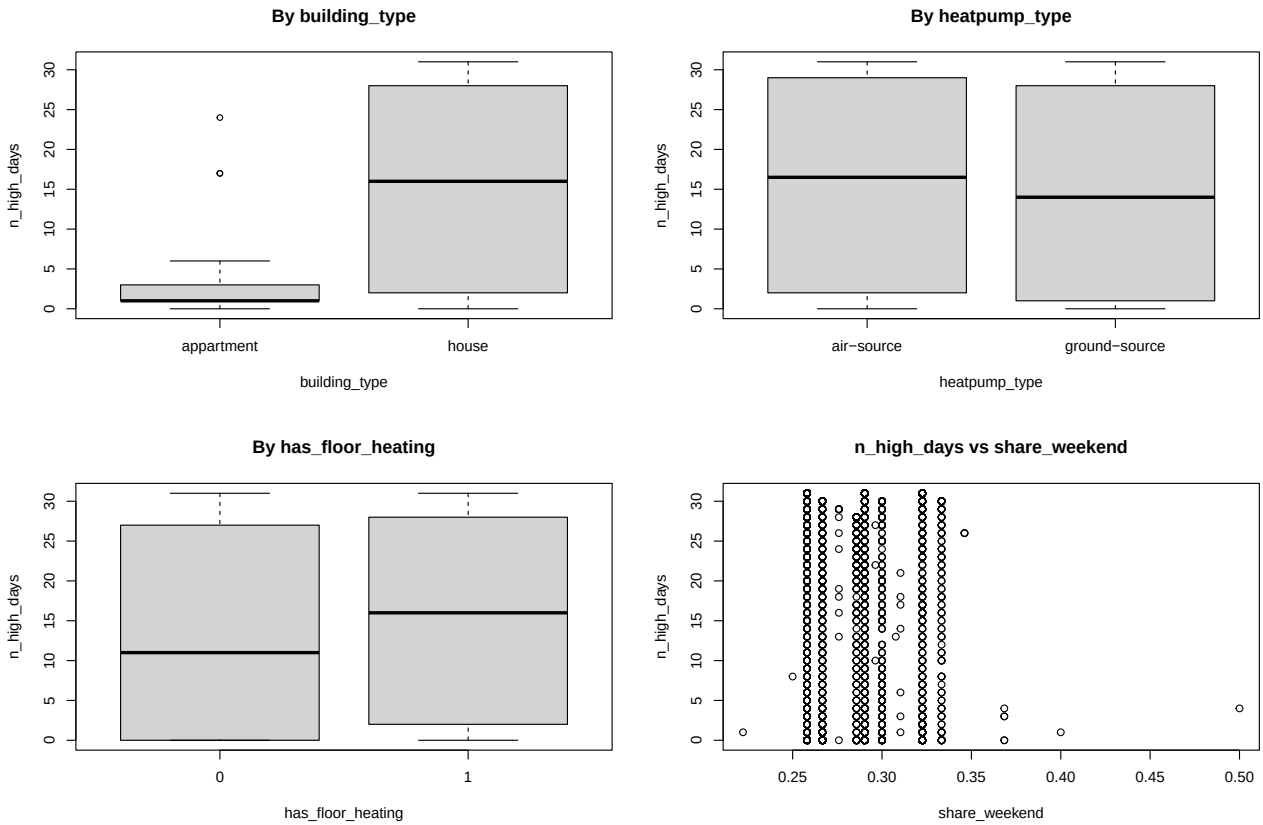
The relationship between `n_high_days` and continuous predictors is explored using scatterplots.

For `heating_degree_days`, a clear positive pattern is visible: higher values are generally associated with higher counts of high-consumption days. This is consistent with increased heating demand during colder periods.

For `temp_avg`, an inverse relationship is observed. Higher temperatures correspond to fewer high-consumption days, while lower temperatures are associated with higher counts, reinforcing the dominant role of weather conditions.

The variables `living_area` and `n_residents` show weaker relationships. While there is a slight tendency for larger households and more residents to be associated with higher counts, the variability is substantial, indicating that these factors alone are not strong predictors.

3.0.2 Categorical and binary predictors



Boxplots are used to compare the distribution of `n_high_days` across groups.

For `building_type`, houses exhibit substantially higher counts and greater variability compared to apartments, suggesting higher energy demand in larger or less efficient buildings.

For `heatpump_type`, differences between air-source and ground-source systems are present but less pronounced, indicating a more moderate effect.

The variable `has_floor_heating` shows a slight shift towards higher counts when floor heating is present, although the distributions overlap considerably.

The variable `share_weekend` does not show a clear relationship with the outcome. The vertical clustering reflects the calendar structure of months rather than a strong behavioural effect, suggesting that weekend share has only a minor influence.

4 4. Poisson Model

4.1 4.1 Specification

A Poisson Generalised Linear Model (GLM) with log link is used to model the number of high-consumption days per household-month. The response variable is `n_high_days`, and the logarithm of the number of observed days (`log(n_days)`) is included as an offset to account for differing exposure across observations.

The model includes weather variables (heating degree days, average temperature), household characteristics (living area, number of residents), and building-related features (building type, heat pump type, floor heating), as well as the share of weekend days.

```
glm_pois <- glm(
  n_high_days ~ heating_degree_days + temp_avg + living_area +
    building_type + heatpump_type + has_floor_heating +
    n_residents + share_weekend,
  family = poisson(link = "log"),
  offset = log(n_days),
  data = dat_count
)
```